Generator headroom and fixed balancing

Scientific finding

Generator-specific policy adjustments are NOT IDENTIFIABLE FROM AVAILABLE DATA. No generator allocation, reserve availability or ramp feasibility is invented. The following conditional aggregate diagnostics are reported separately from implementability.

Generator headroom: identification limits

The final A identifies participant and balanced counter-injection bus components, not generator-specific adjustments. Its exact decomposition is A[:,n]=sigma_n(C_P[:,n]-pi). Thus p(x)=p_background-C_P diag(sigma)(v-x)+pi sum_n sigma_n(v_n-x_n). No net bus injection is treated as generation.

Native generator and interchange Pmin/Pmax and Qmin/Qmax bounds, source buses, reference PV dispatch and security-adjusted background device dispatch are recoverable. The AC evaluator fixes PV P and places all remaining net P in synthetic PQ devices; slack P covers losses. It does not impose generator active-power feasibility on those synthetic injections.

Generator-specific participation factors for A are NOT IDENTIFIABLE FROM AVAILABLE DATA. Policy-dependent generator headroom violations, counts of affected generators, reserve availability, dispatch availability and ramp feasibility therefore cannot be asserted. Native Pmax is a model bound, not independently verified installed/available capacity. Simple Q bounds are not complete Q-V capability curves.

The bus-aggregate table is an explicitly conditional static necessary-condition audit. It separates recovered background generator/boundary dispatch from load/renewable curtailment, then asks whether assigning the entire balanced counter-injection to that same native generator/interchange fleet would exceed aggregate bounds, holding all other components fixed. It does not assign individual devices, change generation or repair policies. It is neither a proof of general physical infeasibility nor a proof of deployability; alternative load/renewable contributions are outside that conditional test. Background boundary totals are frozen in the DC builder, so allowing their native device ranges in this diagnostic is already a relaxation of that construction.

All scheduled policy injections remain the same across retained line/transformer contingencies. No post-contingency generator redispatch, generator outage, frequency control, reserve or ramp model is introduced. Full-year bounds are evaluated on the one-hour grid only. No ramp values are inferred from capacity.

Conditional full-year results

year	policy	Conditional hours	Buses	Up excess MW	Down excess MW
2029	ANNUAL_M1_LP	8760	31	1,265.500	1,230.456
2029	ANNUAL_M2	8760	31	1,220.743	1,204.777
2029	ROLLING_M1	8760	31	1,441.256	1,236.436
2029	ROLLING_M2	8760	31	1,435.205	1,201.256
2029	U	8760	31	1,442.736	1,236.925
2029	C	8760	31	1,441.917	1,204.395
2029	EC	8760	31	1,441.917	1,210.770
2030	ANNUAL_M1_LP	8760	31	1,159.096	1,160.213
2030	ANNUAL_M2	8760	31	1,047.894	1,160.213
2030	ROLLING_M1	8760	31	1,384.845	1,163.735
2030	ROLLING_M2	8760	31	1,313.863	1,163.735
2030	U	8760	31	1,384.845	1,163.735
2030	C	8760	31	1,321.623	1,163.735
2030	EC	8760	31	1,321.623	1,166.686
2031	ANNUAL_M1_LP	8760	37	1,628.294	1,167.806
2031	ANNUAL_M2	8760	37	1,477.533	1,167.806
2031	ROLLING_M1	8760	37	1,877.359	1,174.415
2031	ROLLING_M2	8760	37	1,810.365	1,174.415
2031	U	8760	37	1,810.365	1,174.415
2031	C	8760	37	1,810.365	1,174.415
2031	EC	8760	37	1,818.852	1,185.131

The fleet-only test fails in every hour of every policy-year. This count is structurally driven in part by balancing shares at buses 4, 5, 17, 32, 34, where the reconstructed native generator/interchange fleet contains no device; their combined balancing share is 6.910%. The model also contains separate renewable/load components, so absence from that native fleet does not prove absence of all physical resources. The all-hour failure must not be interpreted as an identified generator violation or as a policy ranking. It demonstrates that the fixed balancing key cannot automatically be interpreted as redispatch of only the inventoried native fleet.

Mathematical scope

Summing Pmin_g<=P_g<=Pmax_g at a bus gives a necessary aggregate inequality. A violation of that aggregate precludes every device allocation within that specified fleet and with those fixed other resource components. The converse does not establish an actual allocation, ramp feasibility or reserve availability. A bus with two generators generally permits many device adjustments with the same bus sum; A cannot distinguish them. Native source bounds are model-defined, not independently authenticated availability or reserve products.

Recovered data and AC treatment

GENERATOR_CAPACITY_INVENTORY.csv records all native device buses, names, service flags, P and Q bounds and the absence of authenticated reserve/ramp information. AC_REFERENCE_GENERATOR_BOUND_CHECK.csv checks the fixed PV reference P against its native bounds; those PV schedules are identical across policies. BACKGROUND_DEVICE_QC.csv binds recovered security-adjusted background dispatch to its cache hash and full physical-case background arrays. Device registry columns distinguish internal generators, fixed signed interchange groups and renewable-downward-flexibility devices. The conditional test never applies P bounds to a net bus injection.

The counter-injection mapping is fixed for normal operation and retained line/transformer states. It is a scheduled-injection assumption, with no generator-outage or corrective-redispatch model. The AC builder holds PV P fixed and creates synthetic PQ injections; its convergence cannot verify active-power feasibility of those unassigned injections. Nonlinear slack P handles losses and is separately recorded, not silently allocated among generators.

Verdict

CONDITIONAL GO for accounting-model interpretation. The conditional bound results must retain their stated fleet-only scope. Generator-level implementability remains unverified and must not be claimed. The historical optimization model, A and annual certificates are unchanged; no headroom-constrained optimum or corrected policy is computed.